

The fix: delete the entry logit, fix the immigration quota

Problem. The previous entry-adjusted population responses (+9–10%) were $\sim 90\%$ driven by an unidentified entry taste scale ($\lambda = 2$) on a nearly saturated margin ($q^* = 0.97$): a free parameter, in utility units, on a knife edge.

Fix. Retire the outside value and logit. Hold fixed the native retention rate $\bar{R} = (1 - s^{out})E_0/B_0 = 0.969$ and the absolute outside inflow $\bar{M} = s^{out}E_0$ (quota rationing, $s^{out} = 0.169$). Solve $SE_0 = \bar{R} S B_0(\text{policy}) + \bar{M}$ jointly with the house price and the balanced budget. Population then responds only through births, with renewal multiplier $(1 - s^{out})/s^{out} = 4.92$, damped by housing scarcity. Baseline is bit-identical; zero recalibration.

Arm	Policy	TFR	Births/HH	Households	$-\Delta\text{Imm. req.}$	Price	Renter ren
Floor	2% tax	+0.38	+0.33	+1.70	1.74	−17.5	−6.
Floor	2% tax + grant	+0.75	+0.67	+3.46	3.52	−17.3	−5.
Tilt	2% tax	+0.20	+0.18	+0.90	0.92	−17.7	−6.
Tilt	2% tax + grant	+0.49	+0.44	+2.23	2.26	−17.2	−5.

All in percent, relative to the funded 1% baseline. “ $-\Delta\text{Imm. req.}$ ” = required immigration at fixed population $\equiv$ percent of the fertility gap closed (same number, different unit). Housing damping is small: arithmetic $4.92 d \ln B_0$ exceeds realized $d \ln S$ by only 0.02–0.10pp. The old logit rows (+9.0–10.1%) survive as a labeled

Labels and open decisions

How to read the population numbers. Long-run stationary comparisons, not forecasts: renewal arithmetic closes half the gap in roughly a century (generational decay $\rho = 0.831$, $g = 26\text{--}30y \Rightarrow 97\text{--}112y$; an age-structured variant with uniform birth timing gives $\sim 230y$). Welfare statements stay per-household; never total-utilitarian across different populations.

Anchor decision (new, from last night's ACS measurement).

Among age-18–22 household entrants, recent immigrants are only 3.7% (all variants 2.9–9.9%) — below the 14.3% feasibility floor. Real immigration sustains the population at *older* arrival ages, so:

1. **Age-aggregated anchor (recommended):** $\bar{M}$ = total net international migration in entrant-household equivalents ($\sim 0.17\text{--}0.22$, brackets the provisional 0.169); age-18 entry is an explicit compression of arrival ages. Results stand as computed.
2. **Literal entry-age anchor:** 3.7% — then the closure is infeasible at every certified arm and requires multi-age immigrant entry or a stronger fertility block.

Fertility-fit stress test. Feasibility requires $B_0/E_0 \geq 0.831$. The repaired-E5 chain-6 winner delivers 0.788 (completed fertility 1.78 vs. 1.918) and is screened out; the floor/tilt arms pass with 3% slack.